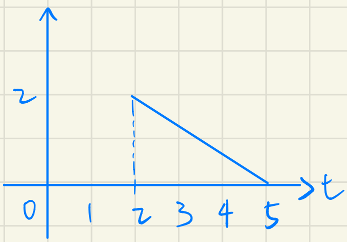
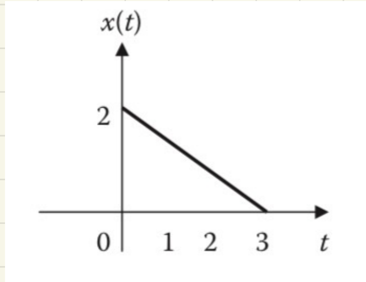
資訊116 F74122048 陳柏亘

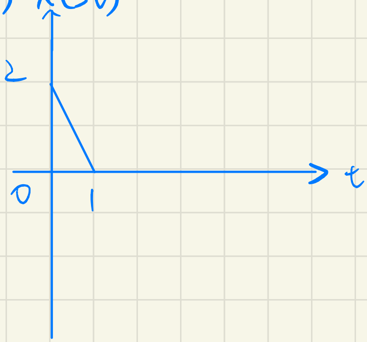
1.24 If $x(t)$ is the signal shown in Figure 1.58, sketch (a) $x(t-2)$, (b) $x(3t)$, and (c) $y(t) = 1 + 2x(t)$.

(a) $x(t-2)$

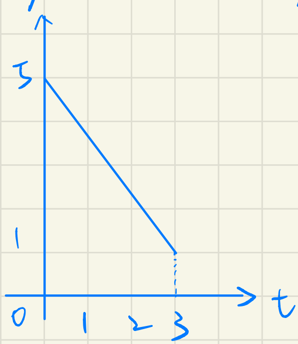




(b) $x(3t)$



(C) $y(t) = 1 + 2x(t)$

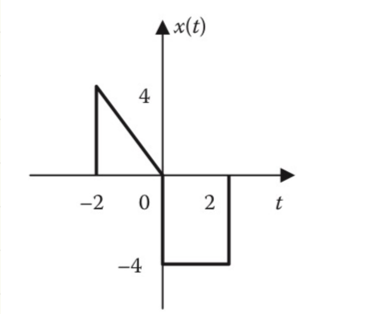


1.29 Given $x(t)$ in Figure 1.59, sketch

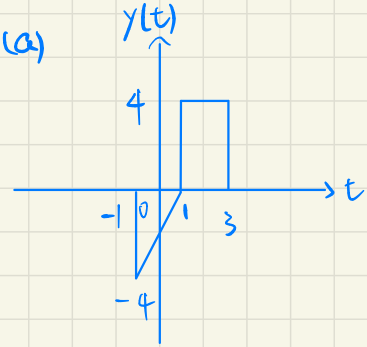
(a) $y(t) = -\,x(t-1)$

(b) $z(t) = 4x(t/2)$

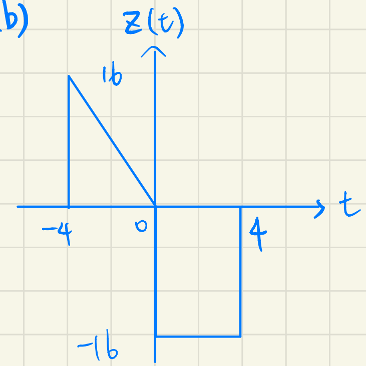
(c) $h(t) = x(2-t)$



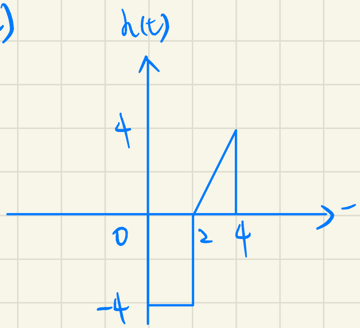
(a)



(b)



(C)

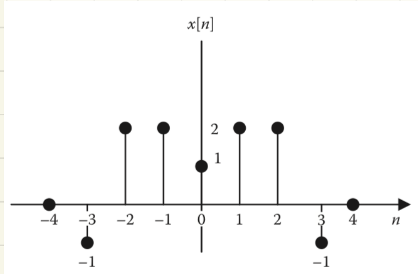


1.32 Consider the discrete-time signal in Figure 1.62. Sketch the following signals:

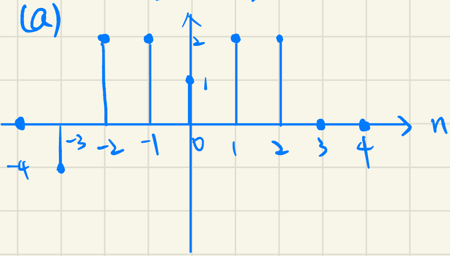
(a) $x[n]u[2-n]$

(b) $x[n]\left[u[n+1]-u[n]\right]$

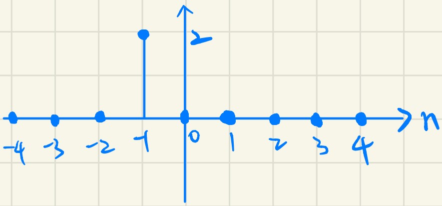
(c) $x[n]\delta[n-2]$



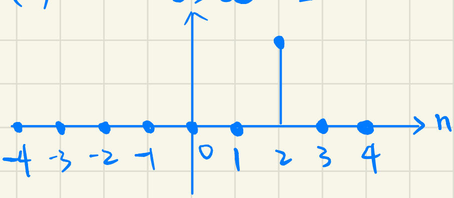
(a) $x[n]u[2-n]$



(b) $x[n][u[n+1]-u[n]]$



(c) $x[n]\delta[n-2]$



1.36 Determine which of the following systems is linear:

(a) $y(t) = \exp\left[x(t)\right]$

(b) $y(t) = \cos x(t)$

(c) $y(t) = t^2 x(t)$

(a) let k_1, k_2 be constants

$x_3(t) = k_1 x_1(t) + k_2 x_2(t)$

$y_3(t) = e^{x_3(t)}$

$= e^{k_1 x_1(t) + k_2 x_2(t)} \neq k_1 e^{x_1(t)} + k_2 e^{x_2(t)}$

$\therefore y(t)$ is non-linear #

(b) let k_1, k_2 be constants

$x_3(t) = k_1 x_1(t) + k_2 x_2(t)$

$y_3(t) = \cos(x_3(t))$

$= \cos(k_1 x_1(t) + k_2 x_2(t)) \neq k_1 \cos(x_1(t)) + k_2 \cos(x_2(t))$

$\therefore y(t)$ is non-linear #

(c) let k_1, k_2 be constants

$x_3(t) = k_1 x_1(t) + k_2 x_2(t)$

$y_3(t) = t^2 x_3(t)$

$= t^2 (k_1 x_1(t) + k_2 x_2(t))$

$= k_1 t^2 x_1(t) + k_2 t^2 x_2(t)$

$\therefore y(t)$ is linear #

1.39 Determine whether the following systems are causal or noncausal, memoryless or with memory.

(a) $y(t) = e^{x(t)}\sin t$

(b) $y(t) = \int_0^t x(\tau)\tau\, d\tau$

(a) $y(t_0)$ only depends on $x(t_0)$

$\rightarrow$ causal and memoryless #

(b) $y(t_0)$ does not depend on future values, but depends on past input

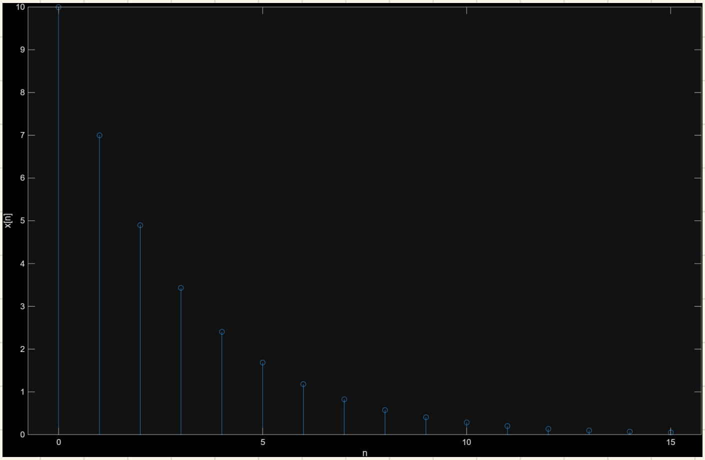
$\therefore$ $y(t)$ is with memory and causal #

1.49 Use MATLAB to plot these discrete-time signals:

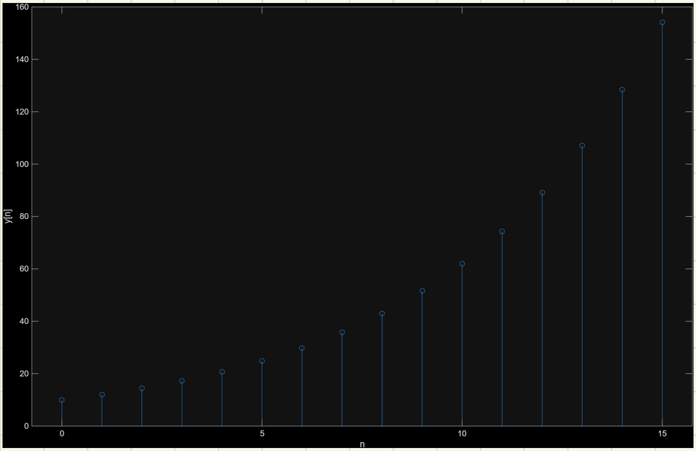
(a) $x[n] = 10(0.7)^n$, $n \geq 0$

(b) $y[n] = 10(1.2)^n$, $n \geq 0$

a.



b.



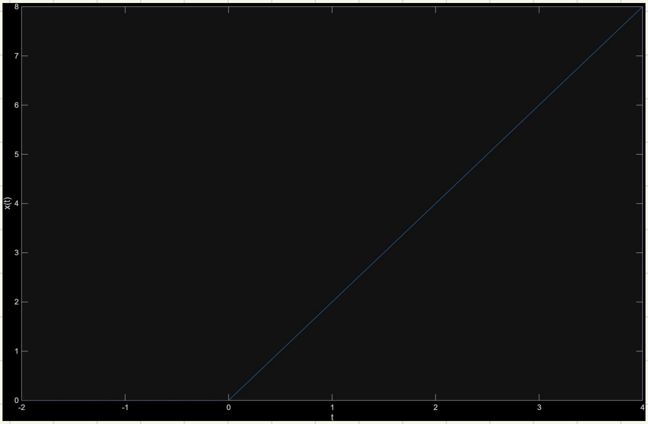
1.50 Use MATLAB to plot the following signals over $-2 \le t \le 4$ s:

(a) $x(t) = 2\, r(t)$

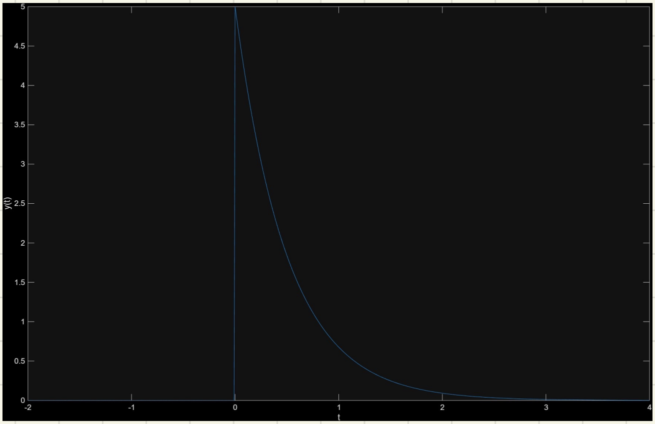
(b) $y(t) = 5e^{-2t}\, u(t)$

(c) $z(t) = 4 \cos 4t + 2 \sin(2t - \pi/4)$

a.



b.



c.